

Session 2

Core Routing Architecture: The Three Critical Routing Tables

In modern computer networking, particularly within dynamic routing protocols such as **OSPF, IS-IS, AODV, and DSDV**, a router does not rely on a single list to pass data. Instead, it systematically builds and maintains **three distinct tables** to discover neighbors, map the network layout, and efficiently forward packets.

1. Neighbor Table

Purpose: Stores vital state information about directly connected neighboring routers that share a common physical or logical link.

Contains:

- Neighbor Router ID (RID)
- Neighbor IP Address
- Local Outgoing Interface
- Link Status / State (e.g., Init, 2-Way, Full)

Example Structure:

Neighbor Router ID	Neighbor IP Address	Local Interface	Link Status
R2	192.168.1.2	Fa0/0	Up / Active
R3	192.168.1.3	Fa0/1	Up / Active

Operational Mechanism:

- The router periodically multicasts/broadcasts **Hello Messages** out of its active interfaces.
- Directly connected neighbors receive these messages and reply back with their own parameters.
- Upon a successful handshake, the router logs the neighbor details into the **Neighbor Table**.
- If no reply or keepalive message is received within a predetermined timeframe (Dead Interval), the neighbor entry is flushed from the table.

2. Routing Table

Purpose: Acts as the primary operational plane data structure, storing only the absolute best, loop-free paths to reach every known destination network.

Contains:

- Destination Network / Subnet Mask
- Next Hop IP Address (or Next Router)
- Metric / Routing Cost (e.g., Bandwidth, Hops)
- Outgoing Interface

Example Structure:

Destination Network	Next Hop	Cost / Metric	Outgoing Interface
10.0.0.0/8	R2 (192.168.1.2)	2	Fa0/0
172.16.0.0/16	R3 (192.168.1.3)	3	Fa0/1

Operational Mechanism:

When a data packet arrives at the router's inbound interface:

1. The router extracts the **Destination IP Address** from the packet header.
2. It performs a lookup against the **Routing Table** (using the longest prefix match rule).
3. It identifies the optimal path, updates the layer-2 header, and forwards the packet directly to the designated **Next Hop** interface.

Example Path: PC1 → R1 → R2 → Server. R1 processes its Routing Table and hands the packet off to R2.

3. Topology Table (Link State Database - LSDB)

Purpose: Maintains a complete, raw architectural map of the entire network structure. It tracks every single router, link, and associated cost within the routing domain. This table is primarily utilized by Link-State routing protocols like **OSPF** and **IS-IS**.

Contains:

- All existing routers in the area / network
- All inter-connecting links and their states
- Calculated link costs
- Full network topology graph

Visual Representation of Network Structure:

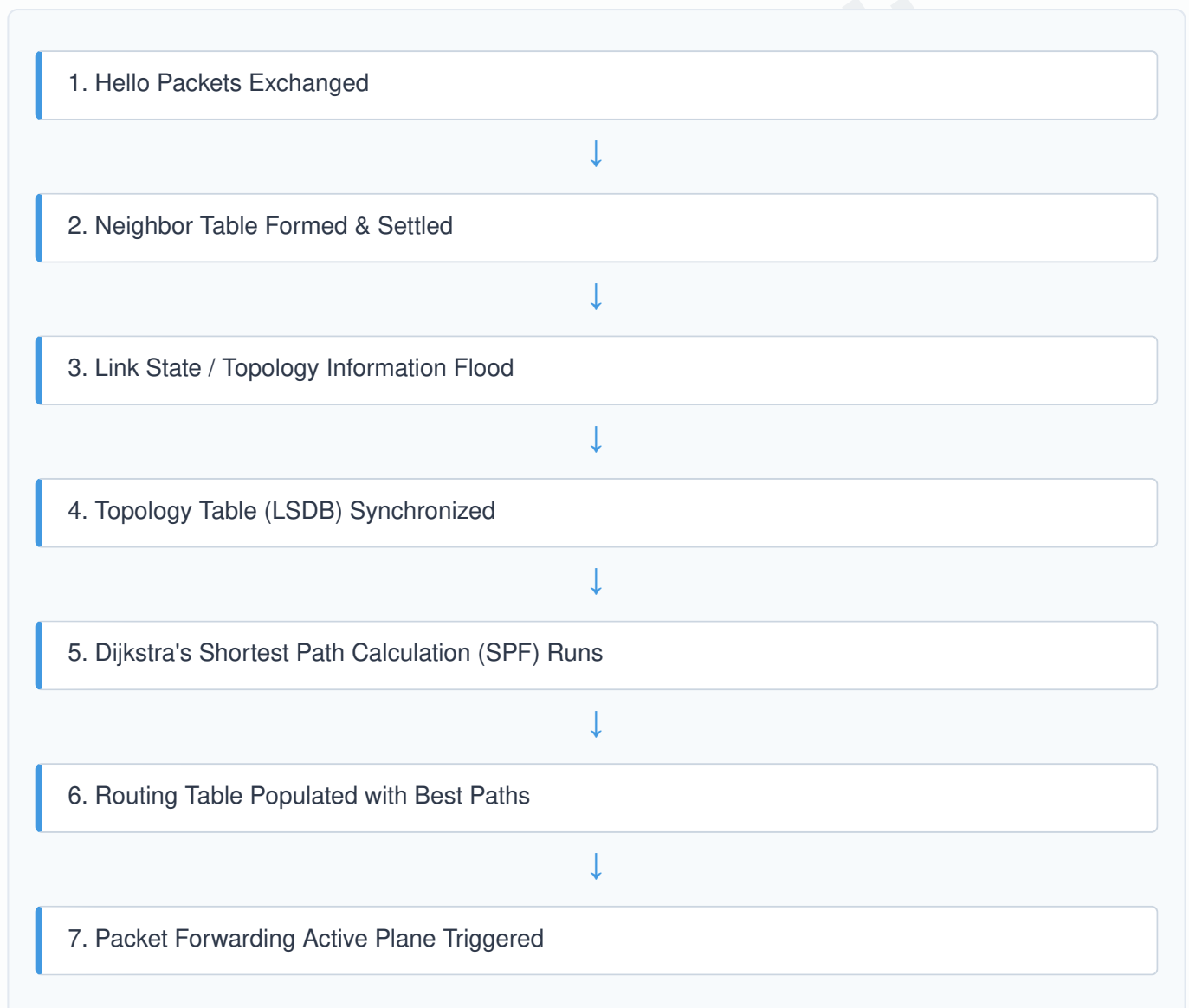
```
graph LR
    R2[R2] --- R1[R1]
    R1 --- R3[R3]
    R1 --- R4[R4]
```

Operational Mechanism:

- Routers reliably flood **Link State Advertisements (LSAs)** or packets containing link info to all peers.
- Every single router aggregates these incoming updates to construct an identical **Topology Table**.
- The router executes **Dijkstra's Shortest Path First (SPF) Algorithm** locally against this topology map.
- The calculated mathematical shortest paths to each network are extracted and copied down into the active **Routing Table**.

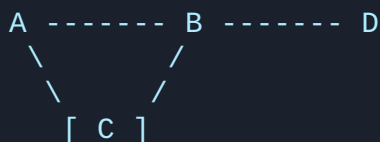
Complete Routing Process Execution

The progression of tables and logic occurs sequentially from structural discovery to data transmission:



Comprehensive Network Scenario

Consider the following structural layout:



Tables Maintained Globally by Router A:

Neighbor Table of Router A:

- Neighbor B
- Neighbor C

Topology Table of Router A:

Maps out the connectivity metrics for segments: A-B, A-C, B-D, C-D, and B-C.

Routing Table of Router A:

Target Destination Network	Next Hop Address
Network B	Directly to B
Network C	Directly to C
Network D	B (Calculated shortest path metric via B)

Data Delivery Sequence: When Router A needs to transmit a payload to destination host **D**, it references its Routing Table, matches target D, identifies **B** as the optimal Next Hop, and shoots the packet out to B. B subsequently forwards it along to D.

Quick Interview Summary

- **Neighbor Table:** Who am I directly plugged into? (Adjacencies)
- **Topology Table:** What does the entire network layout look like? (Map)
- **Routing Table:** Which specific path is the absolute best way to route data right now? (Forwarding)

Workflow Sequence: Neighbor Table → Topology Table → Routing Table → Packet Forwarding